

Advanced Data Analysis Using Python

Time: 3 Hours

Max Marks: 75

Section – A

(Short Answer Questions – Answer any FIVE questions. Each question carries 5 marks)

5 × 5 = 25 Marks

1. What is NumPy? Explain its key features.
2. Write a Python program to create and manipulate a Pandas Series.
3. Explain the difference between `loc[ ]` and `iloc[ ]` in Pandas.
4. What are dataframes in Pandas? List common operations.
5. Describe how to handle missing data in Pandas.
6. Write a note on data visualization using Matplotlib.
7. What is `groupby()` in Pandas? Give an example.
8. Explain the role of Seaborn in data analysis.

Section – B

(Essay Type Questions – Answer ALL questions. Each question carries 10 marks)

5 × 10 = 50 Marks

9. (a) Explain the features of NumPy and demonstrate basic array operations with examples.
OR
(b) Write a program to create multi-dimensional arrays in NumPy and perform slicing and indexing.
10. (a) Discuss in detail the creation, merging, and reshaping of Pandas DataFrames.
OR
(b) Write a Python program to read a CSV file using Pandas and perform basic data analysis.
11. (a) What are the different techniques for data cleaning in Pandas? Explain with code.
OR
(b) Explain how to detect and handle outliers in a dataset using Python.
12. (a) Describe the data visualization techniques using Matplotlib with examples.
OR
(b) Write a Python program to generate line, bar, and histogram plots using Matplotlib.
13. (a) Compare Matplotlib and Seaborn. Explain how Seaborn enhances visualization in Python.
OR
(b) Explain correlation and heatmaps. Write a program to generate a heatmap using Seaborn.